Data Quality Remediation Case Study

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Summary: Reduced issue rate from **55.8%** → **near-zero**, strengthening governance and enabling reliable analytics.
Narrative: Problem → Method → Impact → Value

Problem: An assessment of 500 customer records revealed **319 data quality issues** (**55.8% issue rate**), affecting 279 rows. Invalid Employment Status values dominated, alongside missing income entries and out-of-range credit scores. These gaps created a system-wide validation risk.

Method: A governance-aligned remediation plan was implemented:

- Standardized Employment Status values with controlled taxonomies.
- Normalized credit score ranges (**300–850**).
- Applied median imputation for missing income.
- Capped utilization between **0–1**.
- Mapped free-text entries to standardized codes.

Impact:

- Issues reduced from **319** → **0**.
- Affected rows dropped from **279** → **34**.
- Issue rate fell from **55.8%** → **near-zero**.
- Records flagged as **OK, Corrected (n), or Issue** for traceability.

Value:

- **Reliable Analytics:** Clean data enabled accurate credit risk modeling.
- **Governance & Compliance:** Reduced regulatory and audit risk.
- **Operational Efficiency:** Automated detection minimized manual review.
- **Decision Confidence:** Stakeholders gained trust in dashboards and insights.

This case study demonstrates my ability to **diagnose issues, design governance-aligned solutions, deliver measurable improvements, and translate technical work into business value.**

Raw Data Quality Analysis & Remediation Summary

"Record-level and cell-level data quality assessment"

Payment Behavior

All

500	279	55.80	1.14	Invalid	Employment_Flag
Total Records	Affected Rows	Issue Rate %	Avg Issues / Affected Row	Top Issue Type	Top Issue Field

Problem: Data Quality Issues

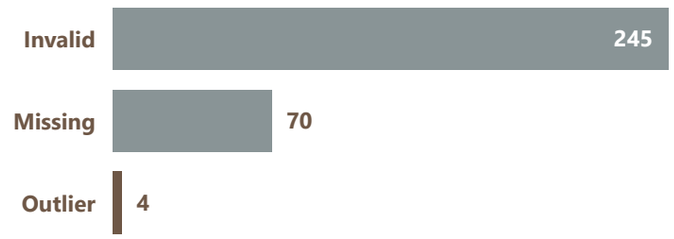
Overall: 319 issues detected across 500 records (55.8% issue rate).
By Issue Type: Invalid values dominate (76.8%), followed by Missing (22%) and Outliers (1%).
By Field: Employment Status (245 issues) is the top driver, followed by Income (39), Loan Balance (29), Utilization (4), and Credit Score (2).
Distribution: Issues are spread across all locations → system-wide validation gap.

Method: Data Quality Remediation

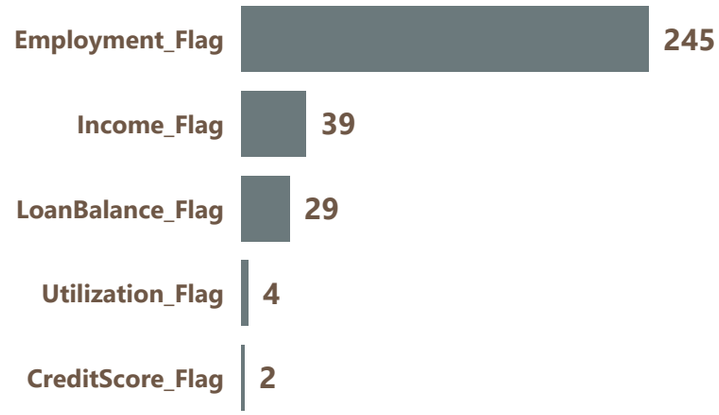
Priority	Issue_Type	Governance_Owner	Remediation_Action
Medium	Contextual	Operations Team	Map free-text entries to standardized location codes
High	Invalid	Data Governance Team	Align with business rules or external reference data
High	Invalid	Credit Bureau Integration	Normalize scoring scale (e.g. 300–850)
High	Invalid	HR/Data Entry Team	Standardize Employment_Status values (use controlled taxonomy)
High	Invalid	HR Team	Validate against realistic age ranges (e.g. 18–100)
Medium	Invalid	Finance/Banking Ops	Standardize card types (Visa, MasterCard, Amex)

Issue Type Distribution

Record-Level & Field-Level Data Quality Assessment



Top Fields Driving Data Quality Issues



Post-Remediation Clean Data Summary

RowQuality
All

319	0	500
Issues (Before)	Issues (After)	Total Record

Affected Rows Breakdown

RowQuality	Count of RowQuality
Corrected (1)	213
Corrected (2)	2
Issue	64
Affected Rows	**279**

Customer_ID	RowQuality	Employment_status_Flag	Income_Flag	LoanBalance_Flag	Age_Flag	Card_F
CUST0001	Corrected (1)	Corrected	OK	OK	OK	OK
CUST0002	OK	OK	OK	OK	OK	OK
CUST0003	OK	OK	OK	OK	OK	OK
CUST0004	OK	OK	OK	OK	OK	OK
CUST0005	OK	OK	OK	OK	OK	OK
CUST0006	OK	OK	OK	OK	OK	OK
CUST0007	Corrected (1)	Corrected	OK	OK	OK	OK
CUST0008	Corrected (1)	Corrected	OK	OK	OK	OK

Post-Remediation Data Quality Impact Summary

Detection: Invalid ages, out-of-range credit scores, inconsistent Employment Status flagged.
Correction: Median imputation, capped utilization (0–1), standardized mappings applied.
Flagging: Records classified as OK, Corrected (n), or Issue.
Impact: 319 issues across 500 records (279 rows). After remediation, issue rate dropped from **55.8%** → **near-zero**, strengthening governance.

Enterprise Value from Remediation

- **Reliable Analytics:** Clean data enables accurate credit risk modeling and reporting.
- **Governance & Compliance:** Strengthened validation rules reduce regulatory and audit risk.
- **Operational Efficiency:** Automated detection and correction minimize manual review effort.
- **Decision Confidence:** Stakeholders can trust dashboards and insights for strategic planning.

Key Takeaway :

This project demonstrates my ability to:

- **Diagnose** complex data quality issues with precision.
- **Design** governance-aligned remediation strategies.
- **Deliver** measurable improvements (issue rate reduced from **55.8% → near-zero**).
- **Translate** technical work into clear business value for analytics, compliance, and decision-making.

Final Note

I approach analytics with **enterprise-grade standards** — ensuring that every dashboard not only visualizes data but also tells a story of **impact, governance, and trust**. My focus is on creating solutions that recruiters and stakeholders can rely on for confident decision-making.